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Neuroimaging Clinical Applications

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Session Objectives

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- Learn the modalities used in radiology to image the brain
- Understand the clinical indications for different types of brain imaging
- Understand contraindications to brain imaging
- How to integrate imaging and other clinical tools in managing neurological disease

Clinical Presentations leading to Neuroimaging

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- Syncope
- Transient Ischemic Attack (TIA)
- Stroke
- Altered Mental Status
- Signs of increased intracranial pressure
  - Papilledema
  - Cushing Triad: widened pulse pressure + bradycardia + irregular respiration
  - Vomiting
  - Headache
- Evaluation of known lesions - tumor, aneurysm, neurologic disease (i.e. multiple sclerosis)

Modalities in Neuroimaging

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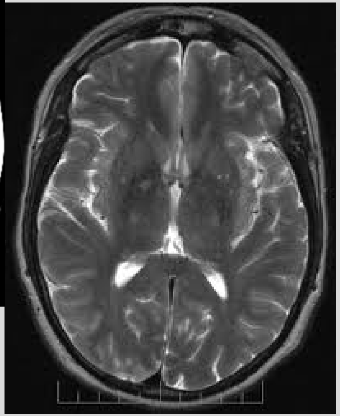
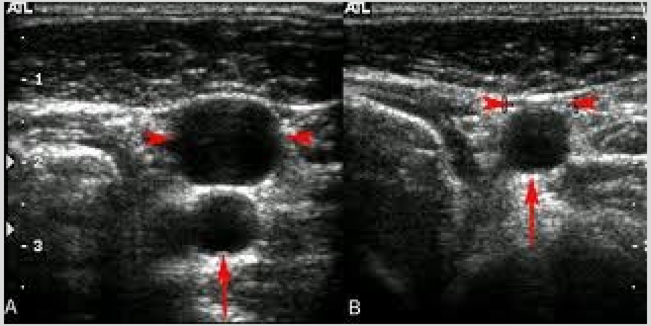
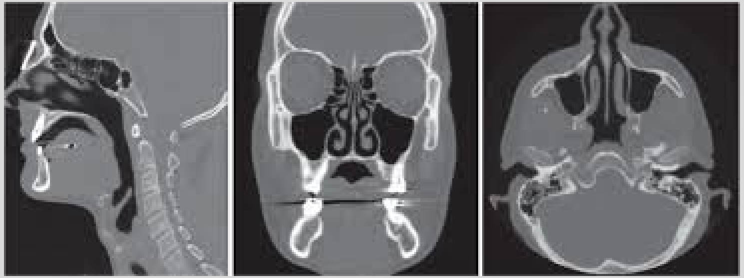
- Cross-sectional Imaging
  - CAT scan (CT)
  - Magnetic Resonance Imaging (MRI)
  - Carotid Ultrasound
- Conventional angiogram





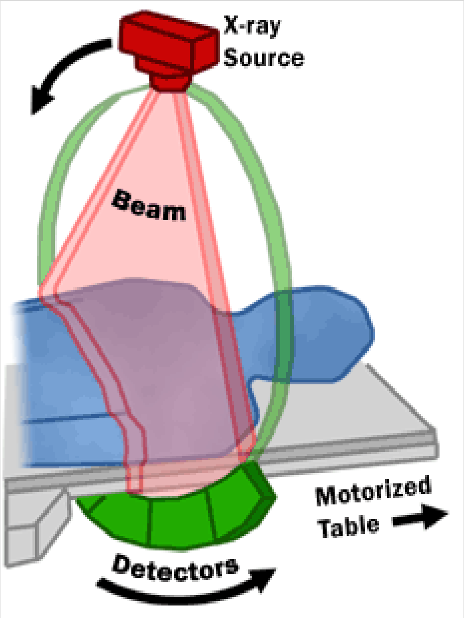
CT Scan

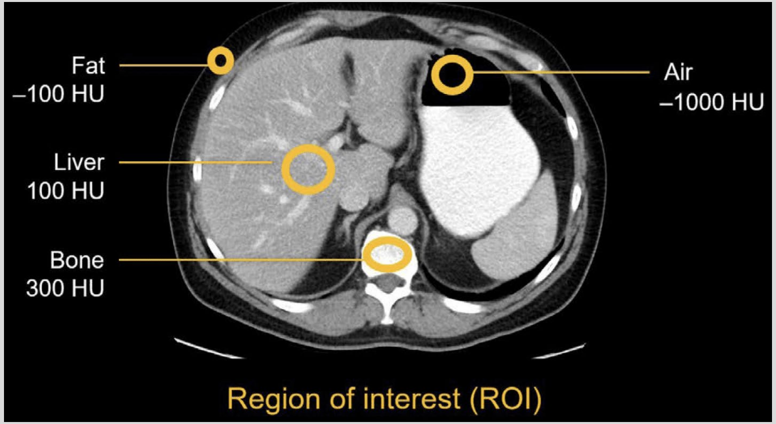
- Helical xray beams are directed through the brain, head, neck
- 2D cross sectional images are compiled, and images may be integrated for a 3D view
- Images are displayed in sagittal, coronal and axial planes



CT

- Each tissue type **attenuates** the xray beams differently, resulting in different **Hounsfield units**





CT Windows

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- CT images are viewed with different “windows” to highlight the attenuation differences of various tissues
  - Bone
  - Soft tissue
  - Lung
  - Brain
  - Subdural
  - Liver

CT: Bone and Brain Windows

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CT: Lung and Soft Tissue Windows

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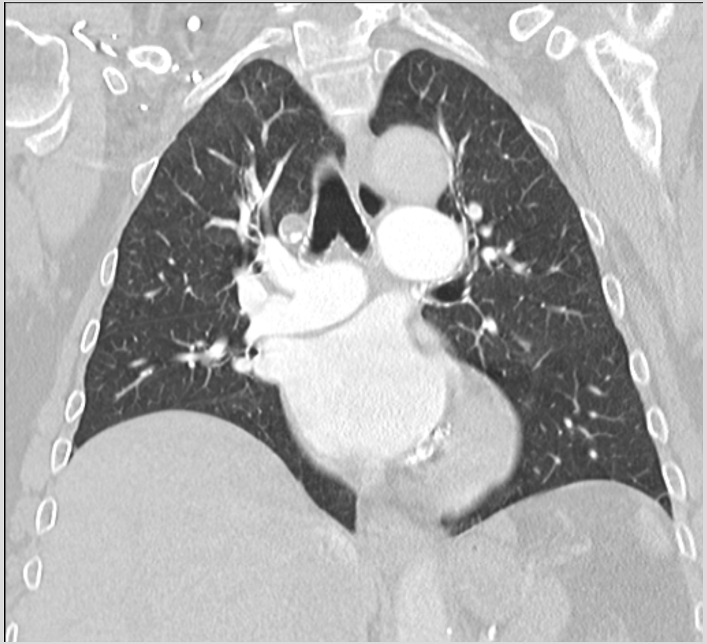

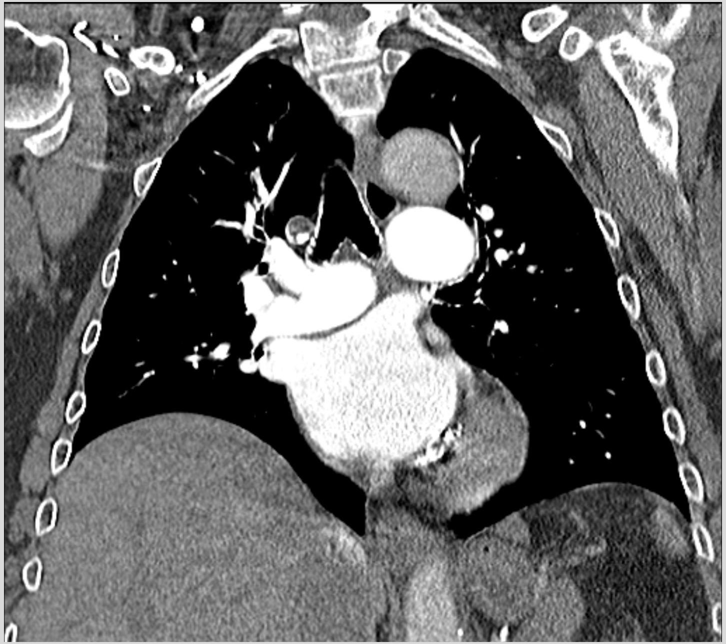

Advantages of CT

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- Superior visualization of:
  - Acute hemorrhage
  - Calcium, Bone
    - Tumor calcifications, fracture
  - Good anatomic detail of multiple tissue types all in one study
- Fast
- Non-invasive
- Relatively cost effective compared to MR

Indications for CT in Neuroimaging

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- Emergencies
  - Trauma
  - Stroke
  - Syncope
  - Altered mental status
- Characterize tumors
- CT Angiography to evaluate arterial rupture, stenosis, aneurysm, dissection

Disadvantage of CT

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ABDOMINAL REGION	Procedure	Approximate effective radiation dose	Comparable to natural background radiation for:
	Computed Tomography (CT)–Abdomen and Pelvis	7.7 mSv	2.6 years
	Computed Tomography (CT)–Abdomen and Pelvis, repeated with and without contrast material	15.4 mSv	5.1 years
	Computed Tomography (CT)–Colonography	6 mSv	2 years
	Intravenous Urography (IVU)	3 mSv	1 year
	Barium Enema (Lower GI X-ray)	6 mSv	2 years
	Upper GI Study with Barium	6 mSv	2 years

BONE	Procedure	Approximate effective radiation dose	Comparable to natural background radiation for:
	Lumbar Spine	1.4 mSv	6 months
	Extremity (hand, foot, etc.) X-ray	Less than 0.001 mSv	Less than 3 hours

CENTRAL NERVOUS SYSTEM	Procedure	Approximate effective radiation dose	Comparable to natural background radiation for:
	Computed Tomography (CT)–Brain	1.6 mSv	7 months
	Computed Tomography (CT)–Brain, repeated with and without contrast material	3.2 mSv	13 months
	Computed Tomography (CT)–Head and Neck	1.2 mSv	5 Months
	Computed Tomography (CT)–Spine	8.8 mSv	3 years

CHEST	Procedure	Approximate effective radiation dose	Comparable to natural background radiation for:
	Computed Tomography (CT)–Chest	6.1 mSv	2 years
	Computed Tomography (CT)–Lung Cancer Screening	1.5 mSv	6 months
	Chest X-ray	0.1 mSv	10 days

DENTAL	Procedure	Approximate effective radiation dose	Comparable to natural background radiation for:
	Dental X-ray	0.005 mSv	1 day
	Panoramic X-ray	0.025 mSv	3 days
	Cone Beam CT	0.18 mSv	22 days

- **Radiation**
  - Current knowledge primarily based on acute *high dose* exposure after the atomic bomb
    - Acute dose of 500 mSV can cause nausea or skin reddening
    - Acute dose of 1000mSV can cause death without treatment
  - **Difficult to quantify risk of *low dose* ionizing radiation on humans – latent period**
    - Normal life (3mSv/yr)
    - Frequent flying (.03 mSv roundtrip NYC-LA)
    - Medical Imaging
    - **Risk of fatal + nonfatal cancer from 1 CT scan is 1 in 2000, 1 in 1000 for nonfatal cancer**
  - **Technical CT scan parameters may be adjusted**
    - **Low dose protocols, protective apron**
  - **Must weigh risks and benefits in pregnancy and children**

CT Contrast

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- An iodinated solution administered to highlight various tissues via three routes: intravenous, oral, rectal
- Oral and rectal forms allow visualization of the GI tract for assessment of patency and perforation
- Intravenous form opacifies the vasculature and is taken up by tissues, in a process termed “enhancement”
- Patterns of enhancement
  - Diffuse/uniform
  - Heterogenous
  - Ring-enhancement

CT Contrast

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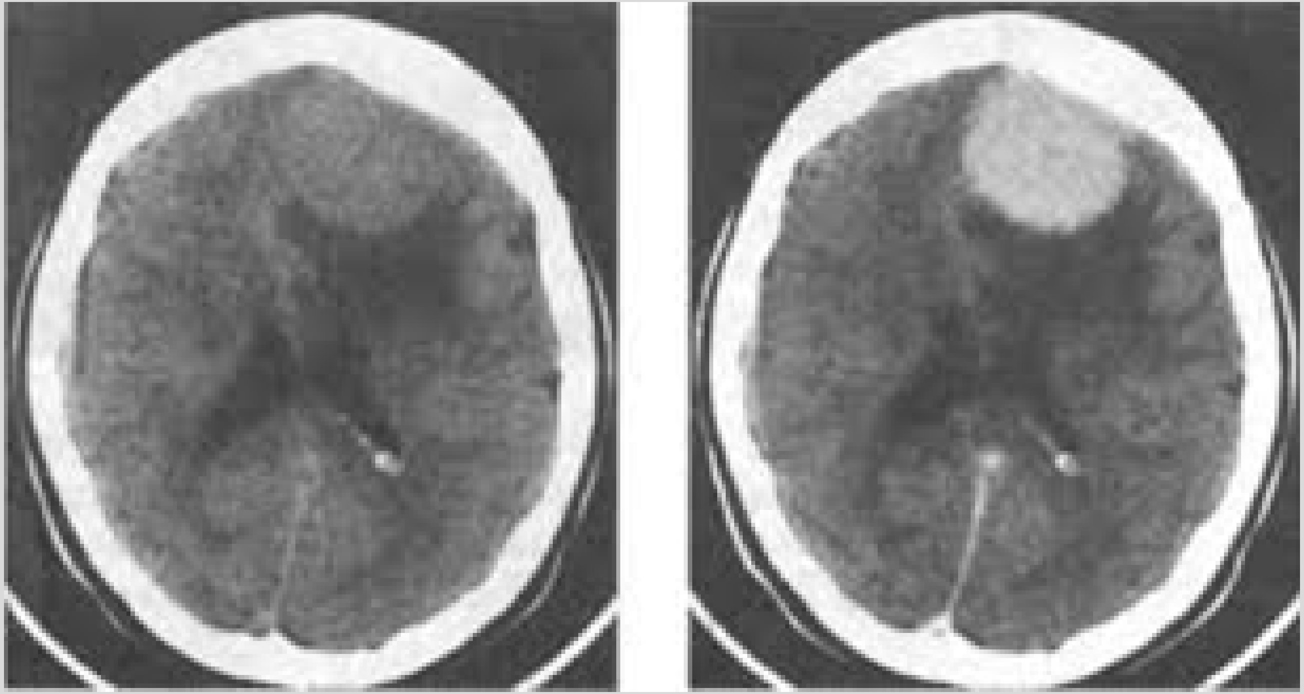

PRE-CONTRAST

POST-CONTRAST

Diffuse Enhancement

Diagnosis - Meningioma

CT Contrast

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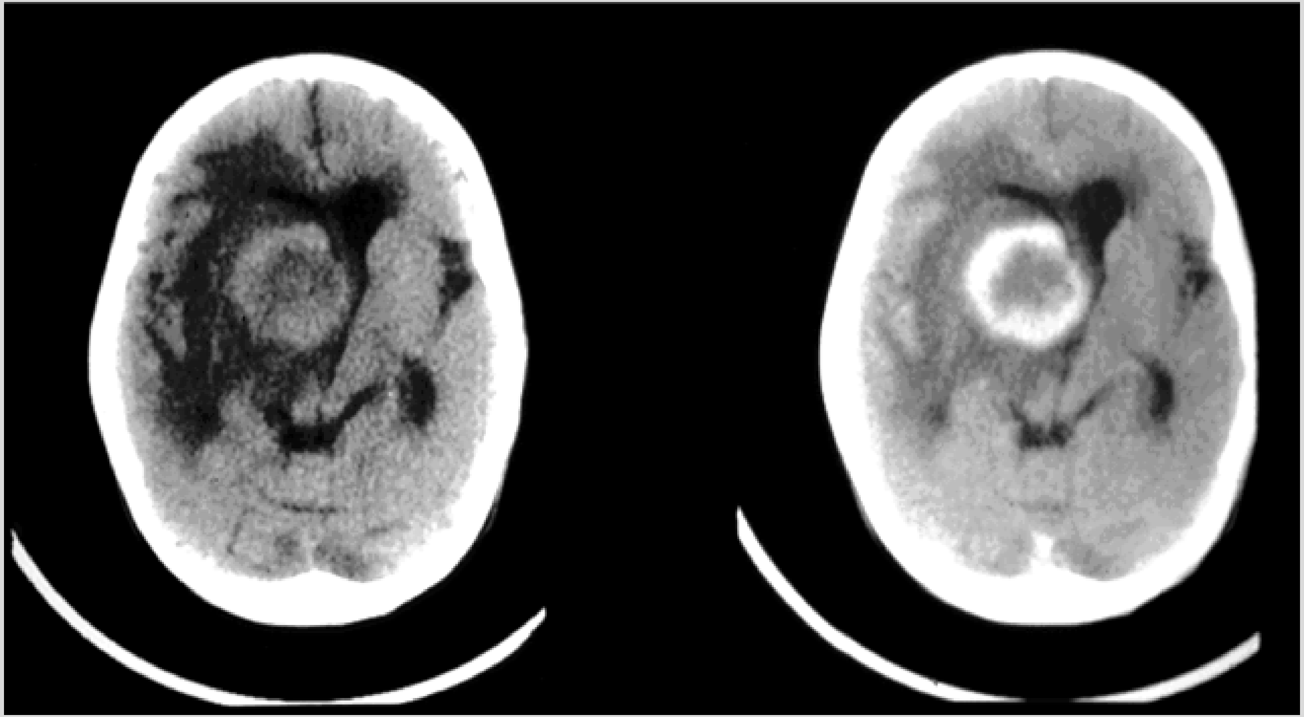

PRE-CONTRAST

POST-CONTRAST

Ring Enhancement

Diagnosis - Abscess

Risks of IV CT Contrast

- Allergic reaction
  - Patients with history of allergy (seasonal, food, previous contrast administration) are prone to having a histamine reaction to contrast
    - Mild, not requiring treatment (5-8%)
      - Feeling of warmth, nausea, itchy throat
    - Moderate (1%)
      - Severe vomiting, hives, swelling
    - Severe (.1%)
      - Life threatening anaphylaxis
  - Management
    - Prescribe contrast with discretion
    - Pre-treatment with Benadryl or Prednisone
    - “crash cart” on site with epinephrine

Risks (Continued)

- Nephropathy
  - Patients at risk:
    - Renal impairment (creatinine >1.5)
    - Diabetics
    - Diabetic on metformin - drug must be discontinued for 48hrs
  - Acute renal failure
    - Pre screen for Cr > 1.5 and other pertinent history
    - Schedule hemodialysis within 24 hrs of contrast administration if patient is on hemodialysis
    - Pre-treat with sodium bicarbonate and hydration
- Excretion into breast milk
- Mortality associated with CT contrast 1 in 1.15 million

Indications for Brain CT with IV Contrast

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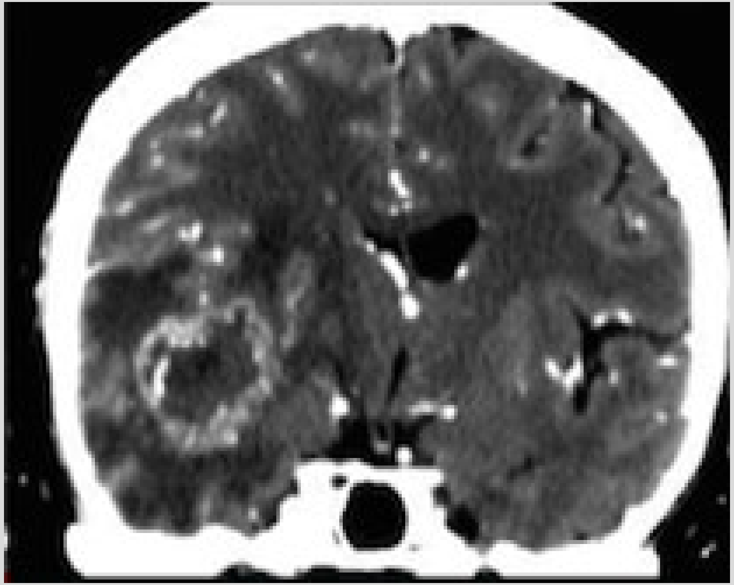
- Suspicion of :
  - Tumor
  - Vascular pathology – stroke, aneurysm, etc.
  - Infection or abscess
- Not used when suspecting acute hemorrhage
  - Acute hemorrhage is visible on CT without contrast
  - Contrast can LIMIT visualization of hemorrhage
  - Free extravasated contrast can irritate brain tissue



Magnetic Resonance Imaging

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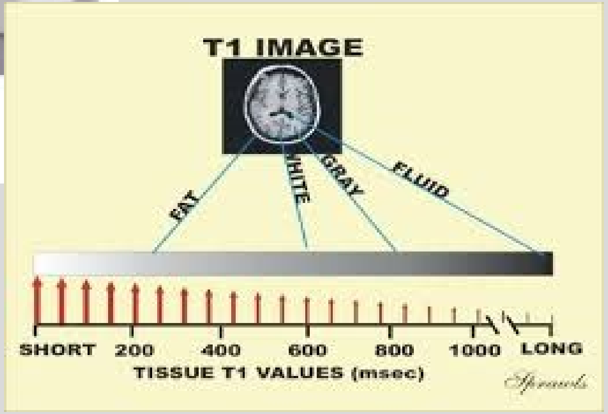
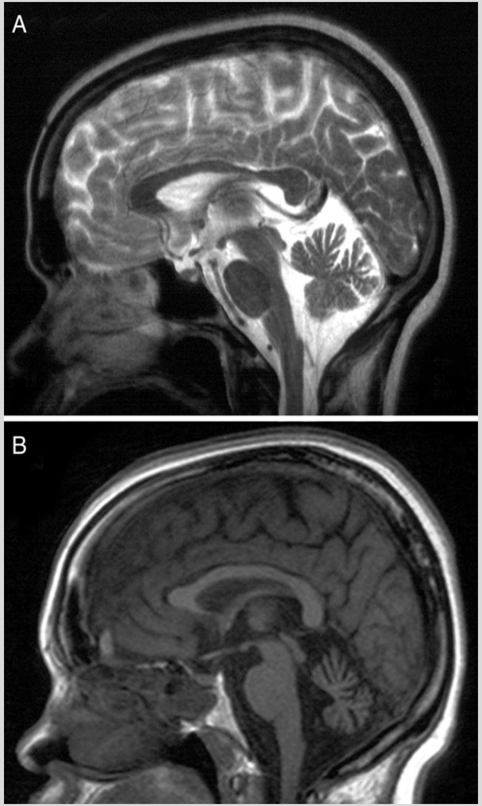
- Magnetic excitation causes changes in alignment of atoms within the body
- Tissues have distinct patterns of proton excitement and relaxation
- Software measures magnetization vectors that result from the relaxation of protons, producing a 3D image



MRI

- Main sequences are T1 and T2, in which magnetization pulses and image acquisition occur at different times
- This difference results in highlighting of different anatomy and pathology
- Additional variations of these sequences include FLAIR, GRE, Diffusion, Proton Density, and increasingly more sequences



T2

T1

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Advantages of MRI

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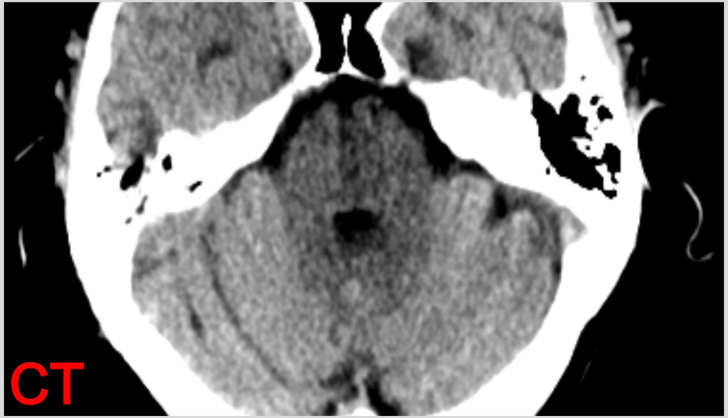
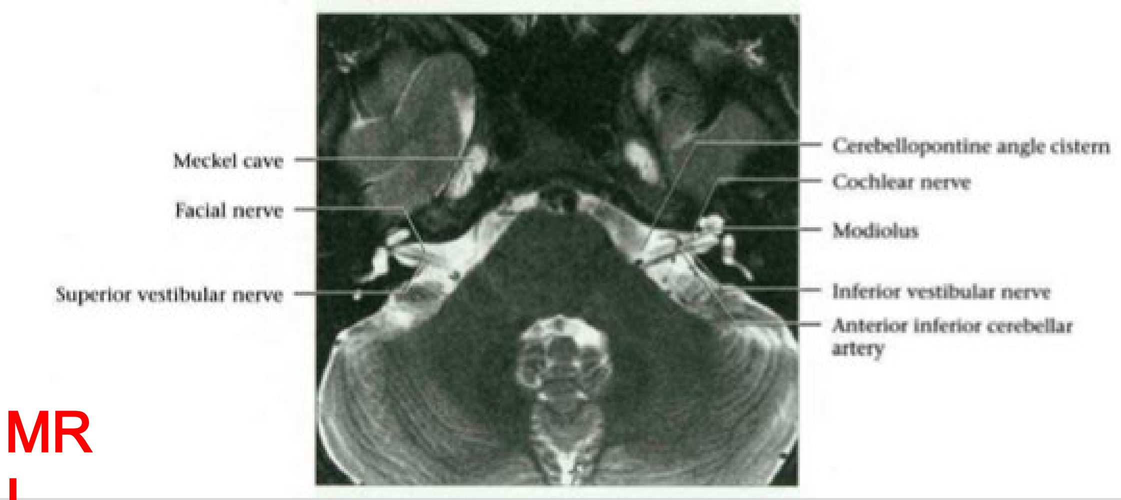
- Superb visualization of soft tissues including brain, spinal cord, nerves, ligaments, tendons and tumors
- No radiation
- Imaging of vasculature in renal compromised patients, without the need for intravenous contrast





Indications for Brain MRI

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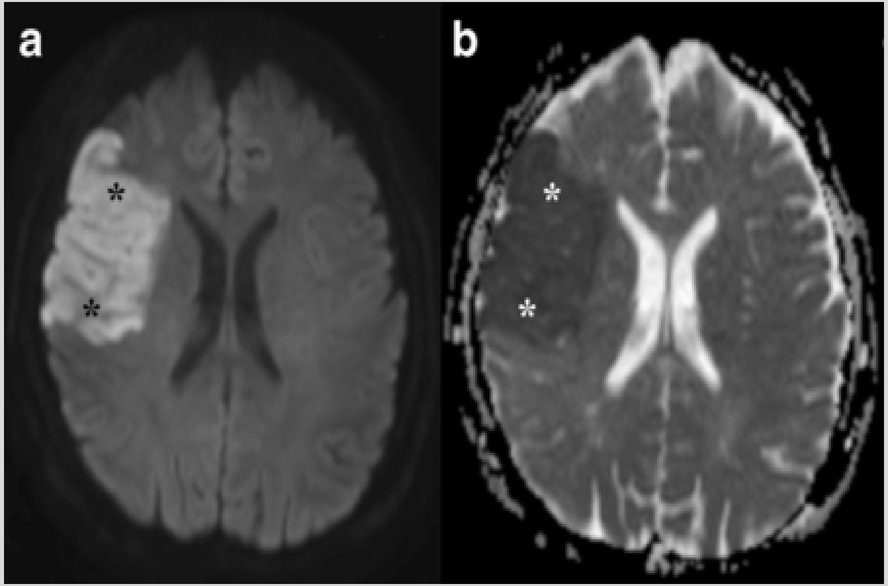
- Early stroke (<12hrs)
  - MRI may be performed after initial CT to detect a CT-occult stroke or to further characterize the extent of ischemia
- Characterization of tumors
- Visualization of cranial nerves
- MR angiography to evaluate stenosis, aneurysm, dissection



Diffusion MRI for stroke

Disadvantages of Brain MRI

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- Costly (up to double CT)
- Claustrophobia
- Slow (approx. 40 min depending on sequences performed)
- Metal implants (pacemaker, old prosthetic heart valves) contraindicated
- Weight limit 350 lb.
- IV contrast risks

MRI IV Contrast

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- Gadolinium - based solution
- Heavy metal group

Indications for MR Brain with IV Contrast

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- Same as for CT:
  - Tumor
  - Vascular pathology
  - Infection

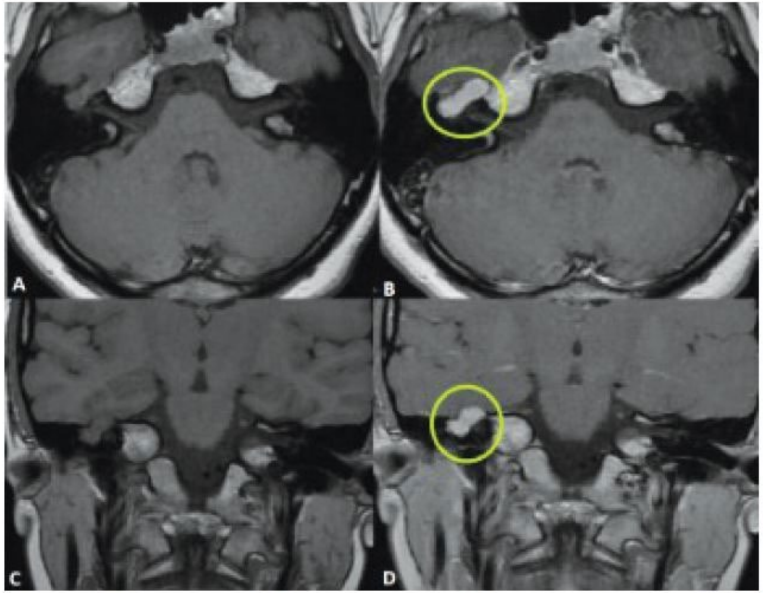

Figure 1. A T1-weighted MRI pre-contrast (A & B) and post contrast (B & D) demonstrate enhancement of the facial nerve schwannoma in an axial (A & B) and coronal (C & D) plane.

MRI Contrast

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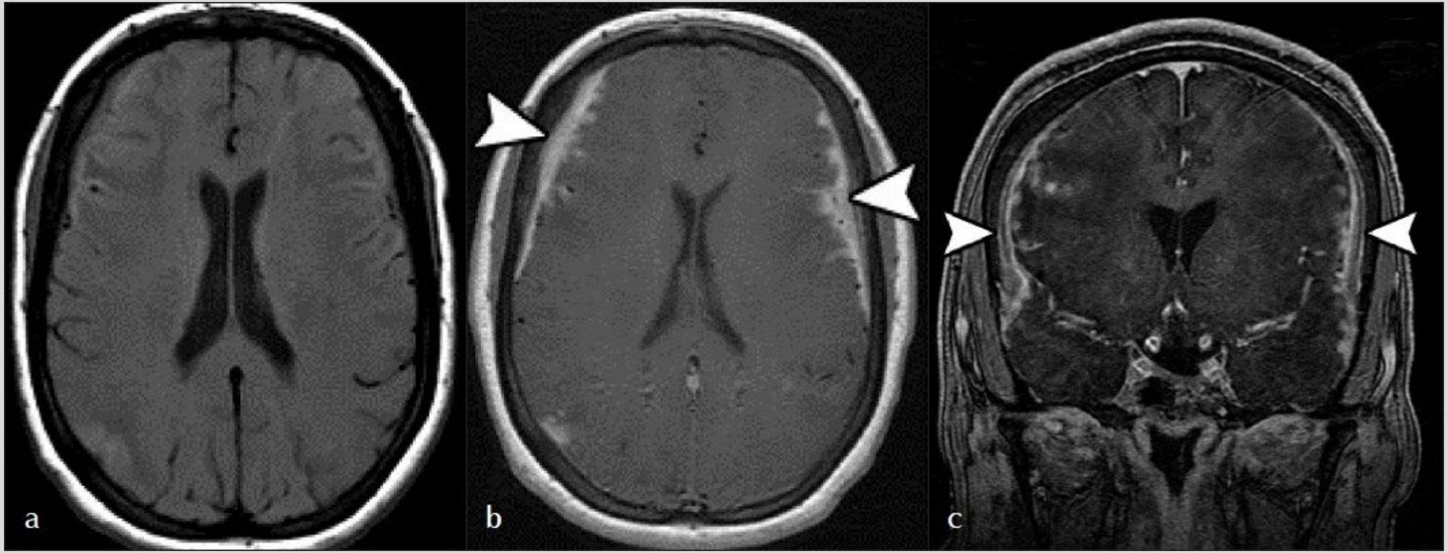

PRE-CONTRAST POST-CONTRAST

Leptomeningeal Enhancement

Diagnosis - Neurosarcoidosis

Risks of MRI IV Contrast

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- Nephrogenic Systemic Fibrosis
  - Rare but serious condition characterized by fibrosis of skin and other tissues which can occur hours to months after IV gadolinium contrast administration
  - At risk:
    - Renal impairment, GFR <30
      - Incidence in this population is about 5%
      - Most institutions do not administer to GFR <30
- Gadolinium deposition disease
- Allergic reaction
  - Hypersensitivity - .35%
  - Mild - 1%
  - Anaphylaxis - .01%

Carotid Duplex Ultrasound

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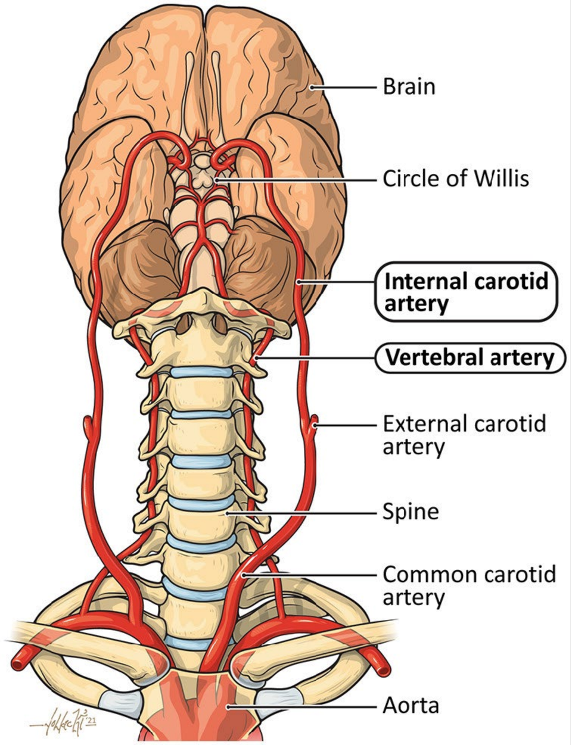
- Cross-sectional imaging of the CERVICAL segments of the major vessels to the head
- Arteries evaluated on a standard duplex ultrasound:
  - Common carotid
  - Internal carotid (cervical segments only)
  - External carotid
  - Vertebral (cervical)
- Veins may also be evaluated



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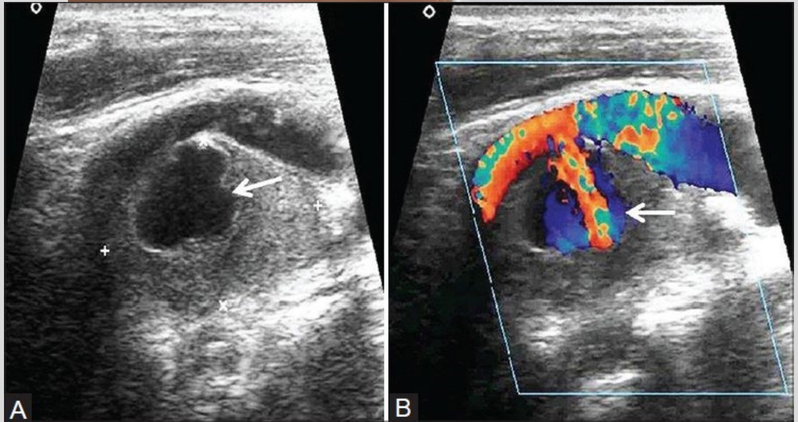
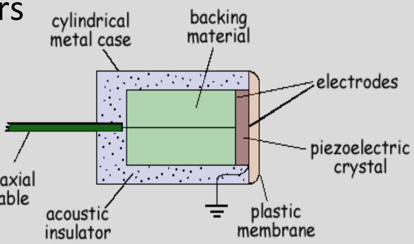
Carotid Duplex Ultrasound

- Step 1 – Grayscale and Color Doppler images
  - Ultrasound transducer is used on the tissue of interest to send and receive sound waves to and from tissue
  - Sound moves differently through various tissues, producing a grayscale image
    - Color doppler - software uses received signal to depict speed and direction of blood flow with colors





Carotid Duplex Ultrasound

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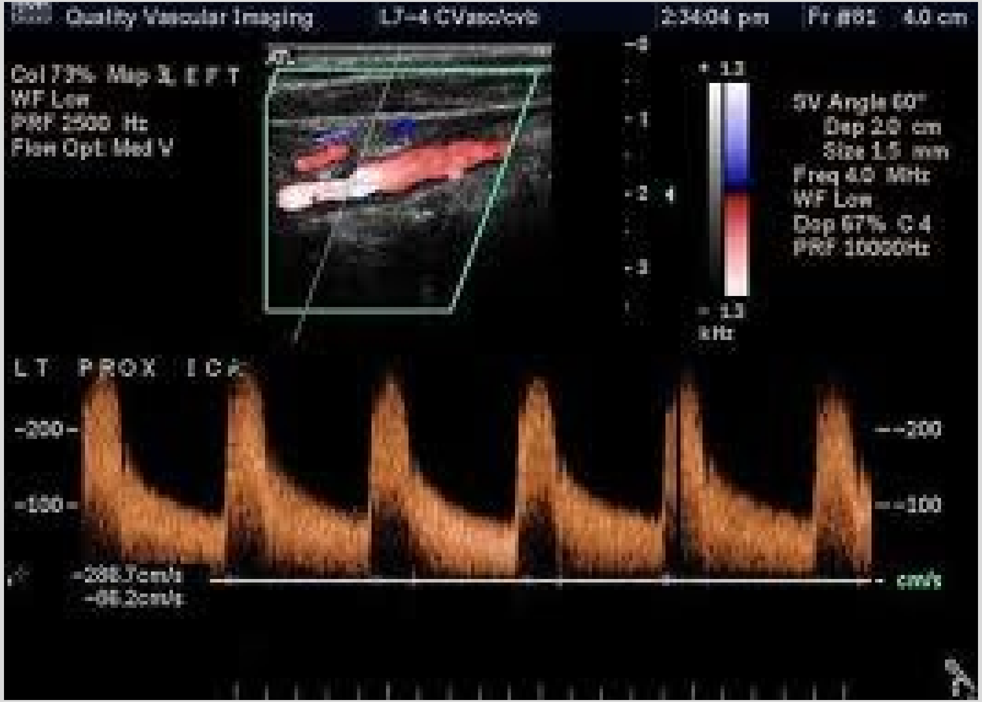
- Step 2 – Spectral analysis
  - Measures blood flow velocity and provides a waveform



Advantages of Carotid Duplex Ultrasound

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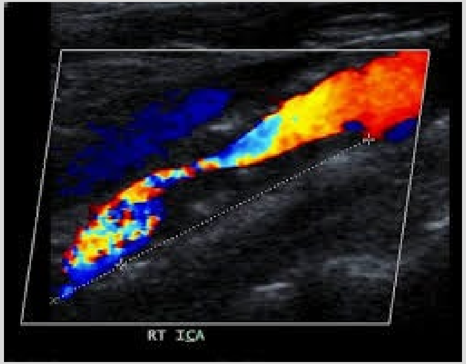
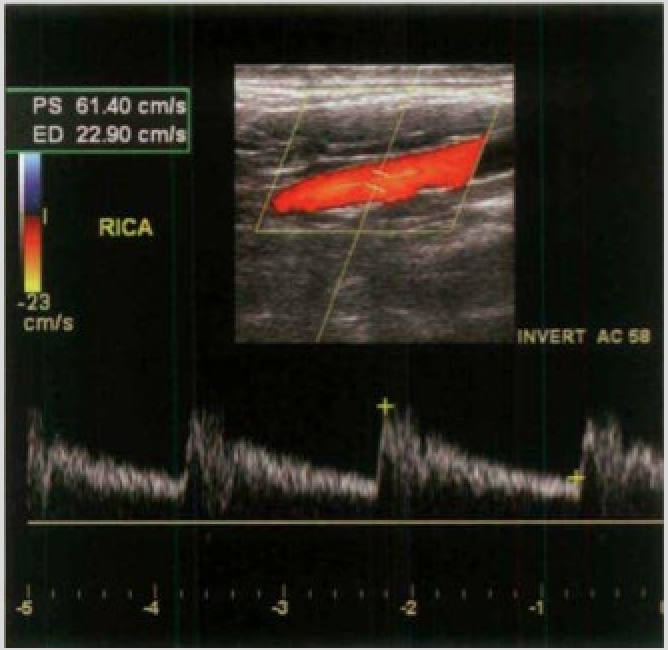
- Non invasive
- Cost effective
- May be performed portably
- Reliable





Indications for Carotid Duplex Ultrasound

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- Suspicion of, or to follow–up a known carotid or vertebral artery stenosis
  - Common Clinical Presentations:
    - Altered mental status
    - Transient ischemic attack
    - Syncope
    - Stroke
    - Bruit on physical exam

Disadvantages of Carotid Duplex Ultrasound

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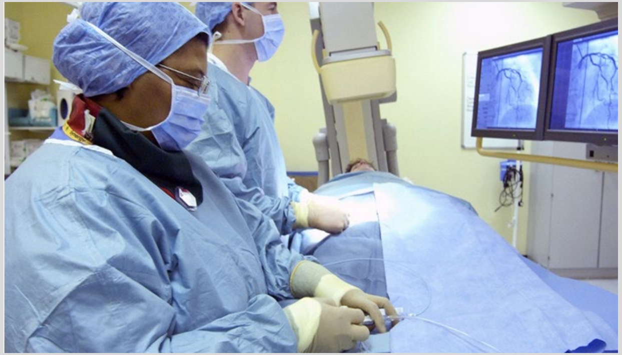
- Technical factors
  - Technician-dependent
  - Image degradation secondary to
    - Large body habitus
    - Patient pain
    - IV lines, ventilator

Conventional Angiogram

- Fluoroscopy-guided imaging of a vessel of interest by accessing a peripheral vein or artery via catheterization and using intravenous contrast
- Performed by neurosurgeons, interventional radiologists, interventional neuroradiologists, interventional cardiologists



Conventional Angiogram

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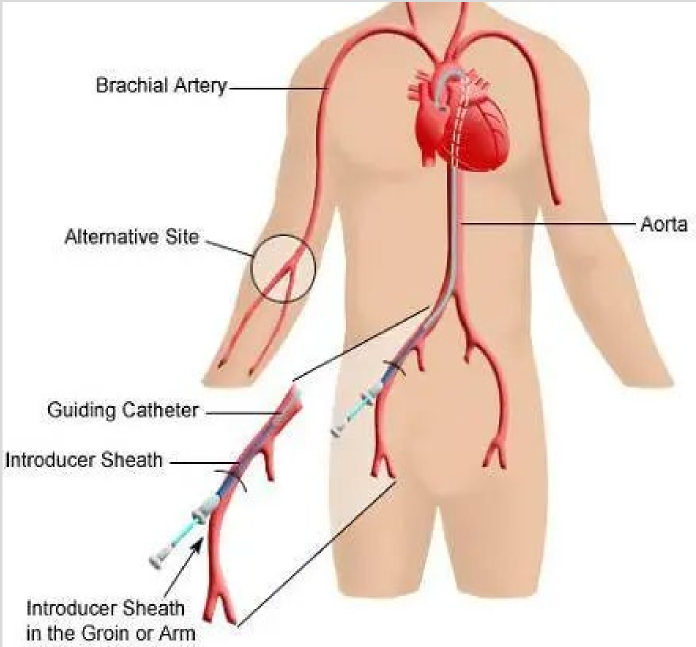
- To visualize or map arteries and veins
  - Identify Pathology
  - Treat pathology



Advantages of Conventional Angiogram

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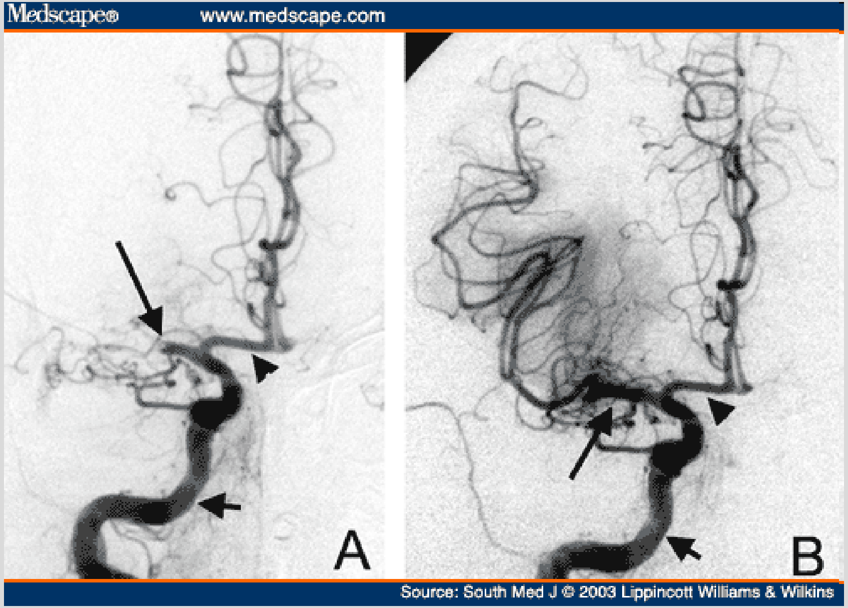
- Diagnostic AND Therapeutic capabilities
  - If pathology is found, can immediately treat
- Superior to CT and MR angiography in evaluating plaque morphology in cases of arterial stenosis



PRE-TREATMENT

POST-TREATMENT

Indications for Conventional Angiogram

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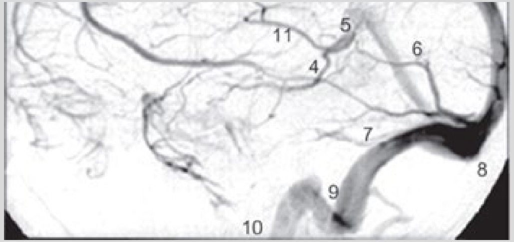
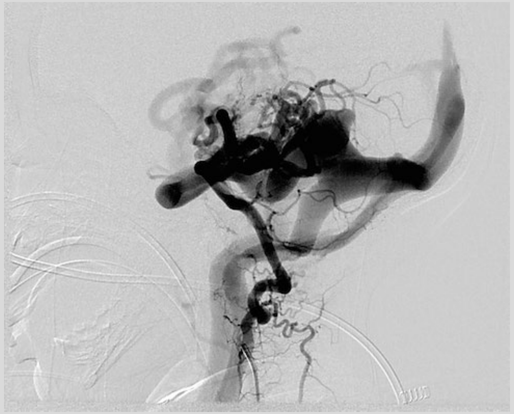
- Stenosis with intention to treat
  - Congenital
  - Atherosclerotic
  - Compression by external organs
  - Angioplasty
    - Balloon inflation within an area of stenosis
    - Often supported by stent placement
    - New * drug-eluting stents
- Traumatic vascular injury
- Aneurysm, arteriovenous malformation
  - Coil embolization
  - Plugs
  - Gel embolization



Normal cerebral venogram



Abnormal cerebral venogram with Diagnosis - arteriovenous malformation

Disadvantages of Conventional Angiogram

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- Invasive
  - Requires sedation
  - Complications of vascular injury, bleeding, infection
  - Risk of IV contrast
  - May require hospital stay
- Limited to visualization of vasculature and clinicians often need additional cross sectional imaging to observe how pathology relates to surrounding organs

Summary

- CT BRAIN
  - Best for acute trauma, intracranial hemorrhage
- MR BRAIN
  - Best for evaluation of nerves, ligaments, and when trying to avoid radiation to the patient
- CT BRAIN/MR BRAIN with INTRAVENOUS CONTRAST
  - To evaluate vascular pathology and patterns of enhancement in other lesions
- CAROTID DUPLEX STUDY
  - Cervical
- CONVENTIONAL ANGIOGRAM
  - Best for vascular pathology and cases in which diagnosis and treatment may be paired
    - Known aneurysm, stenosis

Feedback

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R